



* Practical - 9 *

-> Write a program to implement Travelling Salesman Problem in prolog.

distance C delhi, mumbai, 1400.

distance C delhi, Kolkata, 1500.

distance C delhi, chennai, 2200.

distance C mumbai, Kolkata, 2000.

distance C mumbai, chennai, 1300.

distance C Kolkata, chennai, 1650.

path C X, Y, D :- distance C X, Y, D.

path C X, Y, D :- distance C Y, X, D.

total_distance C [_, _], 0.

total_distance C [A, B | Rest], Total :-

path C A, B, D1,

total_distance C [B | Rest], D2,

Total is D1 + D2.

cities C [delhi, mumbai, Kolkata, chennai].

ts_p C Start, BestFullPath, MinDistance :-

cities C Cities,

member C Start, Cities,

select C Start, Cities, Others,



```
findall (Dist - FullPath ,  
        ( permutation (Others, Perm) ,  
          append ([Start | Perm] ,  
                  [Start] , FullPath ) ,  
          total_distance (FullPath , Dist )  
        ) ,  
        Pairs ) ,
```

```
Keysort (Pairs , Sorted) ,
```

```
Sorted = [MinDistance - BestFullPath | _] .
```

* Output

- ? - tsp (delhi, Path, Distance).

-> Distance = 5850,
Path = [delhi, mumbai, chennai, Kolkata, delhi],